

From Forecasts to Batteries: Grouped Decision-Focused Model Averaging for Capacity Commitment and Storage Siting on Five Continents

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Abstract

Grid operators must commit capacity a day ahead, and planners must decide where to install new batteries. Both decisions rest on how the many available demand forecasts—operators’ own forecasts, statistical models, machine learning and pretrained foundation models—are combined. We propose *grouped decision-focused model averaging* (GDMA), which jointly learns a small number of latent segments of decision units and segment-specific combination weights by minimising the realised decision cost, with each unit’s own cost asymmetry, rather than forecast error; a soft version shrinks each unit’s weights towards its segment and nests pooled weights, unit-specific weights and shrinkage. We prove a finite-sample oracle inequality that separates the cost of learning segment memberships from the cost of learning weights, show that GDMA is asymptotically optimal and attains the risk of the infeasible unit-specific optimum when segments exist, and prove that it recovers the segments with probability approaching one. A planning stage turns the residual risk of the learned rule into an optimal siting of storage. On 125 grid areas in North America, Europe, Oceania, Asia and Africa with an average demand of 1.66 TW, using public hourly data from 2018 to 2026, soft GDMA lowers out-of-sample commitment cost by 17% worldwide (18–20% for the U.S. operators, relative to their official forecasts), outperforms combining the same forecasts for accuracy, and outperforms a decision-focused neural mixture-of-experts gate by nine percentage points. Better combination substitutes for storage: without batteries, soft GDMA is cheaper than the status quo with 60 GW of optimally sited batteries, and it is worth 6–8 GW relative to standard combination rules. Where a storage programme should go depends on the commitment rule; siting on the status-quo error profile loses up to 15% of a small programme’s value.

Keywords: model averaging; decision-focused learning; latent groups; foundation models; capacity commitment; battery storage siting; energy management

1 Introduction

Electricity systems are managed through many local decision units that commit resources before uncertainty resolves. Every day, each of the balancing authorities, national system operators and regional market operators that run the world’s grids must commit enough capacity to meet the next day’s hourly demand. Committing too little forces expensive real-time purchases, the

activation of reserves or, in the worst case, load shedding; committing too much means paying for capacity that sits idle. The cost of a shortfall is not the same everywhere: an operator embedded in a dense web of interconnections can import from its neighbours, while an isolated system cannot. Winter Storm Uri in February 2021, when the weakly interconnected Texas grid shed load for days while better-connected neighbours coped, showed how much this asymmetry matters (Busby et al., 2021). At the same time, governments are committing to large battery-storage programmes, and planners must decide where those batteries should go.

Operators and planners do not lack forecasts. Operators publish their own day-ahead forecasts, simple persistence and calendar rules are always available, statistical and machine-learning models can be trained on the data, and pretrained foundation models now forecast any series zero-shot (Ansari et al., 2024). None dominates everywhere, which makes combining them attractive. How to combine them is less obvious. The forecast-combination and model-averaging literature, from Bates and Granger (1969) to the asymptotic theory of Hansen (2007), Hansen and Racine (2012), Zhang et al. (2013) and Zhang and Liu (2023), offers two options when there are many series. Weights can be estimated series by series, which respects heterogeneity but produces noisy weights when histories are short—the root of the “forecast-combination puzzle” that simple averages often beat estimated weights (Smith and Wallis, 2009; Claeskens et al., 2016). Or one set of weights can be estimated for all series, which is precise but imposes the same rule on systems that face different conditions. Deep learning offers a third option, a network that maps features to weights (Montero-Manso et al., 2020; Gong et al., 2026), at the price of thousands of parameters and weights that are hard to audit. All three typically judge a combination by its statistical accuracy, although the planner is judged by the cost of the resulting decisions; the decision-focused literature in operations management has shown that learning directly for the decision can be much better than forecasting accurately and then optimising (Ban and Rudin, 2019; Bertsimas and Kallus, 2020; Elmachoub and Grigas, 2022).

This paper proposes *grouped decision-focused model averaging* (GDMA) and uses it to answer two management questions: how should grid operators on five continents combine the forecasts they have into day-ahead capacity commitments, and where should new batteries be installed? GDMA assumes that decision units fall into a small number of latent segments whose members are best served by the same combination of candidate rules, and estimates the segments and the segment-specific weights jointly by minimising the realised system-wide decision cost, in money, with each unit’s own cost asymmetry. A soft version shrinks each unit’s own weights towards those of its segment. The number of segments and the shrinkage intensity are chosen by a time-ordered hold-out. Pooled weights, unit-specific weights and shrinkage towards pooled weights are special cases, and the combined commitment is always a convex combination of commitments the operator already understands. A second, planning stage takes the residual shortfall risk left by the learned rule and allocates a storage budget across grid areas.

We make four contributions.

1. **Method.** We formulate the joint learning of latent segments and combination weights under an asymmetric, unit-specific decision loss, give a k -means-type algorithm with regret-space seeding in which every step is a small convex program, and link it to a storage-siting problem that is solved exactly by a greedy algorithm (Section 3). To our knowledge, neither

the model-averaging, the decision-focused learning nor the latent-group panel literature (Bonhomme and Manresa, 2015; Su et al., 2016) has proposed this estimator.

2. **Learning theory.** We prove a finite-sample oracle inequality that separates the cost of learning memberships ($\log G/T$ per unit) from the cost of learning weights ($GM \log(NT)/(NT)$), which formalises why grouping beats unit-specific weights when histories are short; we show that GDMA is asymptotically optimal and, under a segment structure, attains the risk of the infeasible benchmark in which every unit knows its own optimal weights; that it recovers the segments with probability approaching one; and that on this event it coincides exactly with an estimator told the segments (Section 4).
3. **Evidence on five continents.** On 46 U.S. balancing authorities with their official forecasts (Section 6) and on a global panel of 125 grid areas in North America, Europe, Oceania, Asia and Africa with an average demand of 1.66 TW (Section 7), soft GDMA lowers the out-of-sample commitment cost by 16–20% relative to the best single forecaster, beats combining the same forecasts for accuracy, and beats a decision-focused neural mixture-of-experts gate by 9 percentage points worldwide; a pretrained foundation model enters the candidate set on equal terms.
4. **Storage siting.** Better commitment rules substitute for storage: without any batteries, soft GDMA is cheaper than the status quo with 60 GW of optimally sited batteries, and it is worth 6–8 GW relative to standard combination rules. The optimal location of a storage programme depends on the commitment rule, and siting on the status-quo error profile loses up to 15% of a small programme’s value (Section 8).

The paper speaks to the theme of this special issue in both directions it describes: it develops a statistical learning method with guarantees, motivated by a concrete management problem, and it uses new, openly available data and models to answer a planning question of continental scale. Replication code and data-construction scripts are available from the author.

2 Related literature and the research gap

Model averaging and forecast combination. Since Bates and Granger (1969), combining forecasts has been one of the most reliable ways to improve predictive accuracy; Timmermann (2006) and Wang et al. (2023) review five decades of work. Frequentist model averaging chooses weights by minimising an estimate of risk—a Mallows criterion (Hansen, 2007), leave-one-out or K -fold cross-validation (Hansen and Racine, 2012; Zhang and Liu, 2023), or forward validation for time series (Zhang and Zhang, 2023)—and establishes asymptotic optimality, i.e. that the chosen weights attain the lowest attainable risk. The framework now covers dependent data (Zhang et al., 2013), quantile losses (Lu and Su, 2015), general losses (Yu et al., 2024), time-varying weights (Sun et al., 2021), covariate-dependent weights (Gong et al., 2026), covariate shift (Zhang et al., 2026) and prediction intervals (Qu et al., 2025). The forecast-combination puzzle—that equal weights frequently beat estimated weights—is attributed mainly to estimation error in the weights (Smith and Wallis, 2009; Claeskens et al., 2016); remedies shrink estimated weights towards equal weights (Diebold and Shin, 2019) or restrict them (Zou et al.,

2025). When many series must be forecast, weights are either estimated series by series or learned from series features (FFORMA, [Montero-Manso et al., 2020](#)); [Aiolfi and Timmermann \(2006\)](#) cluster *forecasting models* by past performance. All of these approaches either treat each target series separately or impose one rule on all of them, and none groups the *target units*.

Decision-focused learning. Operations research has long recognised that the loss that matters is the decision cost ([Granger, 1969](#)). The data-driven newsvendor literature learns order quantities directly from data ([Levi et al., 2015](#); [Ban and Rudin, 2019](#)); prescriptive analytics maps covariates to decisions ([Bertsimas and Kallus, 2020](#)); and “smart predict-then-optimize” trains predictors on decision regret ([Elmachtoub and Grigas, 2022](#)); see [Mandi et al. \(2024\)](#) for a survey. In energy forecasting, [Wang et al. \(2019\)](#) combine probabilistic load forecasts by minimising pinball loss on the simplex, one series at a time. This literature trains single models or combines forecasts for one unit; it does not exploit cross-sectional structure when many heterogeneous units face related decisions.

Deep learning for forecasting and for combination weights. Pretrained time-series foundation models such as Chronos ([Ansari et al., 2024](#)) now forecast unseen series zero-shot with accuracy close to models trained on the target data, and deep networks are increasingly used to *produce* combination weights: FFORMA maps series features to weights with gradient boosting ([Montero-Manso et al., 2020](#)), and mixture-of-experts model averaging uses a softmax neural gate with covariate-dependent weights ([Gong et al., 2026](#)). These approaches are flexible, but they fit thousands of parameters, their weights are hard to audit, and—except for [Gong et al. \(2026\)](#), who study conditional generative models—they come without optimality guarantees for the combined decision. We include both a foundation model (as a candidate) and a decision-focused neural gate (as a competitor) in our application.

Storage siting. Where to place grid-scale batteries is usually decided with engineering models of the transmission network, solved by decomposition or stochastic programming ([Piansky et al., 2024](#); [Larroux et al., 2026](#)). These models take the uncertainty that storage must absorb as an input. We make that input endogenous: the residual commitment risk depends on how operators combine their forecasts, so the value and the best location of storage depend on the forecasting policy.

Latent group structures. Panel econometrics models unobserved heterogeneity through latent groups estimated by k -means-type algorithms ([Lin and Ng, 2012](#); [Bonhomme and Manresa, 2015](#); [Ando and Bai, 2016](#)) or classifier-Lasso ([Su et al., 2016](#)). These methods group *regression parameters* under squared loss and focus on the parameters themselves, not on combining the outputs of existing forecasting systems under an asymmetric decision loss.

The gap. Table 1 summarises the position of GDMA. No existing method simultaneously (a) combines a library of existing candidate rules, including machine-learning and foundation-model forecasts, (b) chooses weights by the downstream decision cost with unit-specific cost asymmetries, and (c) shares statistical strength across many decision units through latent segments

Table 1: Positioning of GDMA relative to existing approaches.

Approach	Combines candidates	Decision-focused loss	Shares strength across units	Heterogeneous weights
Per-series combination / JMA / forward validation	✓			✓
Pooled combination	✓		✓	
Quantile / general-loss model averaging	✓	✓		✓
Shrinkage towards equal or pooled weights	✓		partial	✓
Feature-based weights (FFORMA)	✓		✓	✓
Neural mixture-of-experts gate	✓	(✓)	✓	✓
Data-driven newsvendor, SPO		✓	✓	
Latent-group panel models			✓	✓
GDMA (this paper)	✓	✓	✓	✓

whose number is chosen from data, with guarantees on optimality and segment recovery; nor does any link the resulting operational risk to the siting of new storage. Filling this gap matters most when windows are short—because systems change and old data are stale—and units are many, which is exactly the situation of system-wide infrastructure planning.

3 Grouped decision-focused model averaging

3.1 The system-wide commitment problem

A planner is responsible for N decision units (in our application the balancing authorities of the U.S. grid). In period t unit i must commit a quantity q_{it} before demand y_{it} is realised. Committing too little costs $u_i > 0$ per unit of shortfall (emergency purchases, reserve activation or, in the limit, curtailed load); committing too much costs $o_i > 0$ per unit of excess (idle capacity that was paid for). The period cost is the newsvendor cost

$$C_i(q, y) = u_i (y - q)^+ + o_i (q - y)^+ = (u_i + o_i) \rho_{\tau_i}(y - q), \quad \tau_i = \frac{u_i}{u_i + o_i}, \quad (1)$$

where $\rho_\tau(r) = r(\tau - \mathbb{1}\{r < 0\})$ is the check function. The system cost that the planner cares about is $\sum_i \sum_t C_i(q_{it}, y_{it})$; under (1) the cost-minimising commitment is the conditional τ_i -quantile of demand (Gneiting, 2011). Critical ratios τ_i differ across units because the consequences of a shortfall differ: an operator that is tightly interconnected with its neighbours can import at moderate cost, while an isolated one cannot.

For every unit a library of M candidate commitment rules is available. Candidate m proposes $q_{it}^{(m)}$, computed from information available before period t ; we write $\mathbf{q}_{it} = (q_{it}^{(1)}, \dots, q_{it}^{(M)})^\top$. The candidates may come from any source: an operator’s own published forecast, simple persistence rules, statistical models or machine-learning models, each turned into a commitment by adding its own τ_i -quantile residual margin. The planner combines them linearly,

$$q_{it}(\mathbf{w}) = \mathbf{q}_{it}^\top \mathbf{w}, \quad \mathbf{w} \in \Delta_M = \left\{ \mathbf{w} \in \mathbb{R}^M : w_m \geq 0, \sum_m w_m = 1 \right\}.$$

The simplex constraint keeps the combined commitment inside the range spanned by the candidates, which is what makes the combination interpretable and auditable by operators, and it

is known to stabilise estimated weights (Zou et al., 2025).

3.2 Three ways to estimate the weights

Let the estimation window contain periods $t = 1, \dots, T$ for which both candidate commitments and realised demand are observed (a forward-validation sample in the terminology of Zhang and Zhang, 2023). Write $\hat{R}_i(\mathbf{w}) = T^{-1} \sum_{t=1}^T C_i(\mathbf{q}_{it}^\top \mathbf{w}, y_{it})$ for the empirical cost of unit i .

Unit-specific weights. $\hat{\mathbf{w}}_i = \arg \min_{\mathbf{w} \in \Delta_M} \hat{R}_i(\mathbf{w})$. This respects heterogeneity but estimates $M - 1$ free parameters from T serially dependent observations per unit; it is the decision-focused counterpart of per-series forecast combination, and suffers from the same estimation noise that underlies the forecast-combination puzzle (Smith and Wallis, 2009; Claeskens et al., 2016).

Pooled weights. $\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \sum_i \hat{R}_i(\mathbf{w})$. This uses all NT observations but forces a single policy on units that may need different ones.

Grouped weights (proposed). We posit that units fall into a small number G of latent segments whose members are well served by the same combination. The GDMA estimator solves

$$(\hat{\gamma}, \hat{\mathbf{W}}) = \arg \min_{\gamma \in \{1, \dots, G\}^N, \mathbf{W} = (\mathbf{w}_1, \dots, \mathbf{w}_G) \in \Delta_M^G} \sum_{i=1}^N \sum_{t=1}^T C_i(\mathbf{q}_{it}^\top \mathbf{w}_{\gamma_i}, y_{it}), \quad (2)$$

and unit i commits $\mathbf{q}_{i,T+1}^\top \hat{\mathbf{w}}_{\hat{\gamma}_i}$. $G = 1$ gives pooled weights and $G = N$ gives unit-specific weights, so the number of segments indexes the whole range between the two extremes. Three features separate (2) from existing proposals. First, the criterion is the decision cost in the planner’s own units (money or energy), so units are grouped by *how they should act*, not by how their forecasts look. Second, memberships and weights are estimated jointly; a two-step procedure that clusters noisy unit-specific weights and then re-estimates is a natural but inferior alternative (Section 5). Third, because each unit keeps its own critical ratio and its own candidate margins, units in one segment share a combination rule but not a commitment level.

3.3 Computation

Problem (2) is a mixed-integer problem, but it has the structure of k -means with a non-Euclidean, data-dependent distortion, as in the grouped fixed-effects estimator of Bonhomme and Manresa (2015). Given memberships, the weights of each segment solve a pooled convex problem on the simplex; given weights, each unit joins the segment whose weights give it the lowest in-window cost. Algorithm 1 alternates the two steps. The weight step is a linear program; for speed we minimise a softplus-smoothed check function, $\rho_\tau^\varepsilon(r) = \tau r + \varepsilon \log(1 + e^{-r/\varepsilon})$, with bandwidth ε equal to 2% of the unit’s typical decision error, by sequential quadratic programming. On our data the smoothed solution is within 0.01% of the exact linear-programming optimum in cost. Each iteration weakly decreases the objective, so the algorithm converges to a local solution in finitely many steps; we run several starts and keep the best. Starting values are drawn by a k -means++ rule (Arthur and Vassilvitskii, 2007) in *regret space*: the next seed unit is drawn with probability proportional to the extra cost that units would incur if forced to use

Algorithm 1 GDMA for a fixed number of segments G

Require: candidate commitments $\{\mathbf{q}_{it}\}$, demand $\{y_{it}\}$, costs $\{u_i, o_i\}$, G , number of starts S

- 1: compute unit-specific weights $\hat{\mathbf{w}}_i$ and the regret matrix $D_{ij} = \hat{R}_i(\hat{\mathbf{w}}_j) - \hat{R}_i(\hat{\mathbf{w}}_i)$
- 2: **for** $s = 1, \dots, S$ **do**
- 3: draw seeds $j_1, \dots, j_G$ by k -means++ on D and set $\mathbf{w}_g \leftarrow \hat{\mathbf{w}}_{j_g}$
- 4: **repeat**
- 5: $\gamma_i \leftarrow \arg \min_g \hat{R}_i(\mathbf{w}_g)$ for all i ▷ assignment step
- 6: re-seed any empty segment with the unit of largest regret
- 7: $\mathbf{w}_g \leftarrow \arg \min_{\mathbf{w} \in \Delta_M} \sum_{i: \gamma_i = g} \hat{R}_i(\mathbf{w})$ for all g ▷ pooled convex step
- 8: **until** memberships do not change
- 9: **end for**
- 10: **return** the solution with the lowest objective (2)

the closest existing seed’s own weights. In rolling applications the memberships of the previous window provide an additional warm start.

3.4 Soft segments

Segments need not be exact. When units within a segment are similar but not identical, a unit may benefit from a rule that sits between its own estimated weights and those of its segment. *Soft GDMA* uses

$$\hat{\mathbf{w}}_i^{\text{soft}} = (1 - \kappa) \hat{\mathbf{w}}_i + \kappa \hat{\mathbf{w}}_{\gamma_i}, \quad \kappa \in [0, 1], \quad (3)$$

where $\hat{\mathbf{w}}_i$ are the unit-specific weights and $\hat{\mathbf{w}}_{\gamma_i}$ the GDMA weights of the unit’s segment. Because the simplex is convex, $\hat{\mathbf{w}}_i^{\text{soft}} \in \Delta_M$. The family nests every benchmark: $\kappa = 1$ is GDMA, $\kappa = 0$ is unit-specific weights, $G = 1$ is shrinkage towards pooled weights (a decision-focused analogue of [Diebold and Shin, 2019](#)), and $G = 1, \kappa = 1$ is pooled weights. The number of segments is chosen exactly as for GDMA (next subsection); given G , κ is chosen on the same hold-out block from $\{0, 0.25, 0.5, 0.75, 1\}$. Choosing G first keeps the search small, which matters because hold-out costs are noisy when windows are short. Segment centres play the role of data-driven prior means, in the spirit of empirical Bayes shrinkage towards group-specific targets.

3.5 Choosing the number of segments

We split the estimation window into an early block (the first two thirds) and a late block. For each $G \in \{1, \dots, G_{\max}\}$ we estimate (2) on the early block and record every unit’s cost on the late block under its estimated membership and weights. Let $G^\dagger$ minimise the total late-block cost. Following the one-standard-error rule ([Hastie et al., 2009](#)), we choose the smallest G whose total late-block cost exceeds that of $G^\dagger$ by no more than one standard error of the summed unit-level cost differences, and then re-estimate on the full window. The rule respects time order and prefers parsimonious segmentations, which are easier to explain to the operators who would have to adopt them.

3.6 From commitment rules to storage siting

The combination rule is an operational decision, re-estimated every week. A planner also faces a slower, capital-intensive decision: where to install new flexible capacity. We link the two with a two-stage design. In the operational stage, the learned rule leaves each unit with hourly shortfalls $s_{it} = (y_{it} - q_{it})^+$ and surpluses $x_{it} = (q_{it} - y_{it})^+$. In the planning stage, a battery with power P_i (MW) and energy $\bar{h}P_i$ (MWh) installed in unit i discharges to cover shortfalls and recharges from committed-but-unused capacity, which has already been paid for. With state of charge e_{it} and round-trip efficiency η , the chronological dispatch is

$$\begin{aligned} d_{it} &= \min\{P_i, s_{it}, e_{it}\}, & c_{it} &= \min\{P_i, x_{it}, (\bar{h}P_i - e_{it} + d_{it})/\eta\}, \\ e_{i,t+1} &= e_{it} - d_{it} + \eta c_{it}, \end{aligned}$$

and every MWh discharged avoids the underage cost u_i . The annual value of battery power, $V_i(P)$, equal to u_i times the expected energy discharged per year, is non-decreasing and typically concave in P , because each additional MW of power is called on only in rarer and larger shortfalls; where the estimated curve is not concave on the grid we replace later increments by the running minimum of the marginal values, which yields a concave lower bound. Given an annualised cost κ per MW of battery power, the planner solves

$$\max_{P_1, \dots, P_N \geq 0} \sum_{i=1}^N \{V_i(P_i) - \kappa P_i\} \quad \text{subject to} \quad \sum_{i=1}^N P_i \leq B, \quad (4)$$

where B is a storage budget or deployment target (for example a national or continental procurement programme). Because each (concavified) V_i is concave, a greedy algorithm that adds power in small increments to the unit with the highest marginal value is optimal for (4); without the budget constraint, each unit simply installs power until its marginal value falls to κ . We estimate V_i on a grid of battery sizes from the out-of-sample shortfall and surplus paths of each commitment rule.

The two stages interact in two ways that matter for management. First, better commitment rules shrink the shortfalls that storage must cover, so part of the storage requirement can be met by better forecast combination instead of new hardware; we measure this *storage equivalent* of better forecasting as the battery power that the status-quo rule needs to reach the cost of the better rule without storage. Second, the siting decision depends on the rule: batteries placed where the status-quo rule is weakest may be placed where they add little once operations improve.

4 Theoretical properties

This section studies the global minimiser of (2) for a given G . We show that (i) its expected decision cost is close to that of the best grouped policy, with an estimation error that separates the price of learning memberships from the price of learning weights; (ii) it is asymptotically optimal and, under a latent segment structure, attains the risk of the infeasible unit-specific oracle; and (iii) it recovers the segments with probability approaching one, in which case its

weights coincide with those of an estimator that is told the segments. Proofs are in Appendix A.

4.1 Setting and assumptions

Let $\mathbf{z}_{it} = (y_{it}, \mathbf{q}_{it})$. Define the population risk of unit i , $R_i(\mathbf{w}) = \mathbb{E} C_i(\mathbf{q}_{it}^\top \mathbf{w}, y_{it})$, and for a grouped policy $(\gamma, \mathbf{W})$ the average risk and its empirical analogue

$$R(\gamma, \mathbf{W}) = \frac{1}{N} \sum_{i=1}^N R_i(\mathbf{w}_{\gamma_i}), \quad \hat{R}(\gamma, \mathbf{W}) = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T C_i(\mathbf{q}_{it}^\top \mathbf{w}_{\gamma_i}, y_{it}).$$

Let $\mathcal{P}_G = \{1, \dots, G\}^N \times \Delta_M^G$ and $R_G^* = \inf_{\mathcal{P}_G} R$.

Assumption 1 (Stationarity and boundedness). *For each i , $\{\mathbf{z}_{it}\}_t$ is strictly stationary, $|y_{it}| \leq B$ and $\|\mathbf{q}_{it}\|_\infty \leq B$ almost surely, and $\max_i(u_i + o_i) \leq \bar{a}$.*

Stationarity refers to the candidate commitments as well as demand; it holds, for instance, when every candidate is computed from a rolling window of fixed length of a stationary process. Boundedness is natural once demand is normalised by a unit's scale. Under Assumption 1 each loss lies in $[0, L]$ with $L = 2\bar{a}B$ and is Lipschitz in the weights: $|C_i(\mathbf{q}^\top \mathbf{w}, y) - C_i(\mathbf{q}^\top \mathbf{w}', y)| \leq \bar{a}B \|\mathbf{w} - \mathbf{w}'\|_1$.

Assumption 2 (Concentration). *There are constants $c > 0$ and an effective sample length $T_e \leq T$ such that, for every fixed $(\gamma, \mathbf{W}) \in \mathcal{P}_G$ and every $x > 0$,*

$$\Pr(|\hat{R}(\gamma, \mathbf{W}) - R(\gamma, \mathbf{W})| > x) \leq 2 \exp(-c N T_e x^2 / L^2),$$

and, for every i and fixed $\mathbf{w}$, $\Pr(|\hat{R}_i(\mathbf{w}) - R_i(\mathbf{w})| > x) \leq 2 \exp(-c T_e x^2 / L^2)$.

Assumption 2 is a high-level condition. It holds with $T_e = T$ when observations are independent (Hoeffding's inequality) and, when units are independent and each $\{\mathbf{z}_{it}\}_t$ is geometrically β -mixing, with $T_e = T/(C \log T)$ by the blocking argument of Yu (1994) or the Bernstein inequality of Merlevède et al. (2011). Cross-sectional dependence through common shocks (weather, in our application) reduces the effective number of independent units; the results continue to hold with N replaced by an effective cross-sectional size.

4.2 An oracle inequality

Theorem 1 (Oracle inequality). *Let Assumptions 1–2 hold and let $(\hat{\gamma}, \hat{\mathbf{W}})$ be a global minimiser of (2). For any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$R(\hat{\gamma}, \hat{\mathbf{W}}) \leq R_G^* + 2L \sqrt{\frac{GM \log(3NT_e) + N \log G + \log(2/\delta)}{c N T_e}} + \frac{4\bar{a}B}{N T_e}. \quad (5)$$

The square-root term has a transparent decomposition. Per unit, learning the segment membership costs $\log G/T_e$, while learning the segment weights costs $GM \log(NT_e)/(NT_e)$. The same argument applied to unit-specific weights ($G = N$) gives $M \log(NT_e)/T_e$, and applied to pooled weights ($G = 1$) gives $M \log(NT_e)/(NT_e)$ but a larger approximation error $R_1^* - R_N^*$. Grouping therefore replaces the M per-unit parameters of unit-specific weights by a single

discrete label that costs $\log G$ rather than $M \log(NT_e)$ to learn. This is the formal sense in which GDMA resolves the tension between the noise of unit-specific weights and the bias of pooled weights: with $M = 7$ candidates, $N = 46$ units, $G = 4$ segments and an effective length of $T_e = 28$ (one observation per day of a four-week window), the variance proxy under the square root is 0.23 for GDMA against 2.07 for unit-specific weights, so the bound is about three times tighter, while the approximation error of GDMA is zero whenever the segment structure is correct.

4.3 Asymptotic optimality

Following the model-averaging literature (Hansen, 2007; Hansen and Racine, 2012; Zhang and Liu, 2023), call a weight choice asymptotically optimal if the ratio of its risk to the lowest attainable risk in its class tends to one.

Corollary 1 (Asymptotic optimality). *Let the assumptions of Theorem 1 hold, with G and M possibly growing, and suppose $R_G^* \geq \kappa > 0$. If $GM \log(NT_e)/(NT_e) \rightarrow 0$ and $\log G/T_e \rightarrow 0$, then $R(\hat{\gamma}, \hat{\mathbf{W}})/R_G^* \rightarrow_p 1$. If, in addition, there are memberships γ^0 with $G_0 \leq G$ segments such that $\arg \min_{\mathbf{w}} R_i(\mathbf{w})$ depends on i only through γ_i^0 , then $R_G^* = R_N^* = N^{-1} \sum_i \min_{\mathbf{w}} R_i(\mathbf{w})$ and*

$$\frac{R(\hat{\gamma}, \hat{\mathbf{W}})}{N^{-1} \sum_i \min_{\mathbf{w} \in \Delta_M} R_i(\mathbf{w})} \rightarrow_p 1.$$

The second statement is the relevant one for practice: under a segment structure, GDMA attains the risk of the infeasible benchmark in which every unit knows its own optimal weights, although it estimates only $G(M - 1)$ continuous parameters. Unit-specific weights attain the same limit only if $M \log(NT_e)/T_e \rightarrow 0$, a much stronger requirement when windows are short because the system is non-stationary and old data are of limited use. The condition $R_G^* \geq \kappa$ excludes the degenerate case of perfect foresight, which never occurs in commitment problems.

4.4 Recovery of the segments

Assumption 3 (Segment structure). (a) *There are $\gamma^0 \in \{1, \dots, G\}^N$ and distinct $\mathbf{w}_1^0, \dots, \mathbf{w}_G^0 \in \Delta_M$ such that, for all i and all $\mathbf{w} \in \Delta_M$, $R_i(\mathbf{w}) - R_i(\mathbf{w}_{\gamma_i^0}^0) \geq \lambda \|\mathbf{w} - \mathbf{w}_{\gamma_i^0}^0\|_2^2$ for some $\lambda > 0$.* (b) *$\min_{g \neq h} \|\mathbf{w}_g^0 - \mathbf{w}_h^0\|_2 \geq d > 0$ and every segment contains at least πN units, $\pi > 0$.*

Part (a) is a quadratic-growth condition. It holds when the conditional density of demand given the candidates is bounded away from zero near the optimal commitment and the candidates are not collinear, because the Hessian of R_i is then $(u_i + o_i) \mathbb{E}[f_{y|q}(\mathbf{q}^\top \mathbf{w}) \mathbf{q} \mathbf{q}^\top]$, which is positive definite on the tangent space of the simplex. Part (b) requires segments to be genuinely different and non-negligible, as in Bonhomme and Manresa (2015).

Theorem 2 (Segment recovery and oracle equivalence). *Let Assumptions 1–3 hold with G equal to the true number of segments, and let $N, T_e \rightarrow \infty$ with $\log N/T_e \rightarrow 0$. Then, up to a relabelling of segments, (i) $\max_g \|\hat{\mathbf{w}}_g - \mathbf{w}_g^0\|_2 \rightarrow_p 0$; (ii) $\Pr(\hat{\gamma} \neq \gamma^0) \leq N C_M \exp(-c' T_e) + o(1) \rightarrow 0$ for constants $C_M, c' > 0$ that do not depend on N or T ; (iii) with probability approaching one, $\hat{\mathbf{W}}$ equals the infeasible estimator $\tilde{\mathbf{W}} = \arg \min_{\mathbf{W}} \hat{R}(\gamma^0, \mathbf{W})$ that knows the segments.*

Part (iii) is an exact equivalence, not an asymptotic approximation: on the event that memberships are recovered, the joint minimiser and the known-segment minimiser are the same object. Any distributional result available for the known-segment estimator therefore transfers to GDMA without modification. The recovery condition $\log N/T_e \rightarrow 0$ is mild: the number of units may grow polynomially in the window length.

Soft GDMA. For fixed (G, κ) , R_i is convex in $\mathbf{w}$, so the risk of (3) satisfies $R_i(\hat{\mathbf{w}}_i^{\text{soft}}) \leq (1 - \kappa)R_i(\hat{\mathbf{w}}_i) + \kappa R_i(\hat{\mathbf{w}}_{\gamma_i})$. Averaging over units, the excess risk of soft GDMA is at most the κ -weighted average of the excess risks of unit-specific weights and of GDMA, each controlled by Theorem 1. Under a segment structure, therefore, any κ bounded away from zero inherits the faster rate of GDMA for the κ share of the combination, while $\kappa < 1$ buys robustness when segments are only approximate. Choosing (G, κ) from a finite grid by hold-out adds only a $\log(|\mathcal{G}|)$ term to the hold-out error, where $|\mathcal{G}|$ is the grid size.

5 Monte Carlo evidence

5.1 Design

We simulate $N \in \{50, 200\}$ units and $M = 5$ candidate commitment rules. Unit i has demand $y_{it} = \mu_i + x_{it}$, with $\mu_i \sim U[20, 100]$ and x_{it} a stationary AR(1) process with coefficient 0.5 and standard deviation 5, and critical ratio $\tau_i \sim U[0.6, 0.95]$ ($u_i = \tau_i/(1 - \tau_i)$, $o_i = 1$). Candidate m forecasts $f_{itm} = y_{it} - e_{itm}$, where the errors follow AR(1) processes with coefficient 0.5 and innovations with unit-specific scales s_{im} and skewness signs $k_{im} \in \{-1, 0, +1\}$ (a centred gamma distribution for ± 1 , a normal distribution for 0). Each candidate becomes a commitment by adding the τ_i -quantile of its errors in a pre-sample of 200 periods. Three designs define the heterogeneity: *grouped*—every unit belongs to one of three segments with segment-specific error profiles (s_i, k_i) , so that different candidates are best in different segments; *homogeneous*—all units share the first profile; *continuous*—each unit’s error scales are a Dirichlet(1,1,1) mixture of the three profiles, so that no two units share their optimal weights and GDMA’s segment structure is misspecified. Weights are estimated on $T \in \{48, 168, 672\}$ periods and evaluated on 1,000 new periods. We report the out-of-sample cost relative to an infeasible oracle that uses, for every unit, the exact cost-minimising weights on the evaluation sample itself. Each cell uses 50 replications.

5.2 Results

Table 2 supports the theory and clarifies when grouping pays.

Soft GDMA is the most reliable method. It has the lowest cost, or is within 0.1 percentage points of the lowest, in every one of the 18 cells. Its advantage is largest when windows are short: with true segments and $T = 48$ it is 3.5–4.1% above the oracle against 8.4–8.6% for unit-specific weights, and even under continuous heterogeneity, where segments are misspecified, it beats unit-specific weights (6.2–6.8% against 9.1–9.4% above the oracle) and shrinkage towards pooled weights (7.5%). When units share one error profile it matches pooled weights for short

Table 2: Monte Carlo results: out-of-sample newsvendor cost relative to the unit-specific oracle (means over 50 replications), average selected number of segments $\bar{G}$ of GDMA, and adjusted Rand index (ARI; Hubert and Arabie, 1985) between estimated and true segments for GDMA and two-step clustering. Bold: lowest in the row.

Design	N	T	Out-of-sample cost / oracle							$\bar{G}$	ARI	
			EW	Pooled	Per-unit	Shrink	2-step	GDMA	Soft		GDMA	2-step
Grouped	50	48	1.496	1.414	1.084	1.090	1.055	1.050	1.041	3.4	0.87	0.75
	50	168	1.491	1.410	1.033	1.033	1.031	1.030	1.023	3.7	0.88	0.79
	50	672	1.481	1.398	1.013	1.013	1.019	1.018	1.011	5.3	0.68	0.67
	200	48	1.491	1.418	1.086	1.089	1.049	1.040	1.035	3.1	0.93	0.79
	200	168	1.492	1.419	1.032	1.032	1.028	1.023	1.019	5.2	0.71	0.73
	200	672	1.491	1.416	1.012	1.012	1.021	1.017	1.011	5.8	0.62	0.66
Homogeneous	50	48	1.321	1.027	1.097	1.027	1.032	1.028	1.028	1.1	—	—
	50	168	1.322	1.025	1.036	1.020	1.026	1.024	1.021	1.3	—	—
	50	672	1.322	1.025	1.015	1.013	1.017	1.016	1.012	2.6	—	—
	200	48	1.321	1.026	1.094	1.025	1.027	1.026	1.025	1.0	—	—
	200	168	1.320	1.025	1.038	1.020	1.026	1.024	1.020	1.3	—	—
	200	672	1.321	1.025	1.014	1.013	1.017	1.015	1.012	2.9	—	—
Continuous	50	48	1.206	1.148	1.091	1.075	1.090	1.091	1.068	2.7	—	—
	50	168	1.206	1.148	1.035	1.036	1.051	1.055	1.031	4.0	—	—
	50	672	1.203	1.145	1.014	1.014	1.035	1.035	1.014	5.4	—	—
	200	48	1.202	1.146	1.094	1.075	1.090	1.082	1.062	3.1	—	—
	200	168	1.205	1.149	1.035	1.038	1.050	1.049	1.030	4.9	—	—
	200	672	1.206	1.150	1.013	1.013	1.040	1.037	1.014	5.9	—	—

windows and beats them for long ones, where the heterogeneity in critical ratios makes some unit-level adjustment worthwhile.

Hard segments trade bias for variance exactly as Theorem 1 predicts. Plain GDMA beats unit-specific weights in 96–100% of the replications with true segments and $T = 48$, and in all replications with homogeneous units for $T \leq 168$. With long windows ($T = 672$), unit-specific weights are estimated precisely and become slightly better (1.2–1.3% against 1.7–1.8% above the oracle). Here GDMA selects five to six segments rather than three, because units within a true segment have different critical ratios and therefore different optimal weights: with enough data the hold-out rule correctly detects this finer structure, which is also why the ARI falls from about 0.9 at $T = 48$ to about 0.65 at $T = 672$. Soft GDMA removes this residual bias through unit-level shrinkage and is never worse than unit-specific weights by more than 0.1 percentage points.

Joint estimation recovers segments better than clustering estimated weights. With short windows, the ARI of GDMA is 0.87–0.93, against 0.75–0.79 for two-step k -means on noisy unit-level weights, and GDMA’s cost is lower in every grouped cell with $T \leq 168$. When units share one error profile, the one-standard-error rule selects a single segment in most replications with $T \leq 168$ ($\bar{G} = 1.0$ –1.3).

6 Day-ahead capacity commitment on the U.S. grid

6.1 Data

We use the complete public record of the U.S. Energy Information Administration’s Hourly Electric Grid Monitor (Form EIA-930; [U.S. Energy Information Administration, 2026](#)), downloaded on 29 September 2026. For every balancing authority (BA) in the lower 48 states the file reports hourly demand, the BA’s own day-ahead demand forecast and total interchange with neighbouring BAs. We use the period 1 January 2019 to 26 September 2026 in UTC hours and exclude the fourteen regional aggregates. Following [Ruggles et al. \(2020\)](#), non-positive values and values outside $[0.3, 3]$ times a centred seven-day rolling median are treated as missing, and gaps of up to three hours are interpolated. We keep BAs with at least 97% valid demand, at least 93% valid official forecasts, mean demand above 50 MW and a median absolute official forecast error below 15% of median demand. Forty-six BAs remain (Table 12 in Appendix C); they range from PJM (mean demand 92.5 GW) to the City of Homestead (71 MW), include all seven independent system operators, and account for 97% of lower-48 demand. Ten BAs are dropped, mostly small systems with long gaps or implausible official forecasts; 0.27% of the retained BA-hours are missing.

6.2 Decision problem and cost calibration

Before each operating day d , BA i commits hourly capacity $q_{i,d,h}$, $h = 0, \dots, 23$. The information set contains realised demand up to the end of day $d-1$ and the BA’s published day-ahead forecast for day d . Excess commitment costs $o_i = 1$ per MWh, which normalises all costs to units of the per-MWh cost of idle committed capacity. Shortfalls cost $u_i = \tau_i / (1 - \tau_i)$, where the critical ratio τ_i reflects the BA’s ability to import. We measure interchange flexibility in the pre-sample year 2019 as the range between the 5th and 95th percentiles of hourly total interchange, divided by mean demand, and let τ_i decline linearly in the flexibility rank from 0.95 for the least flexible BA to 0.80 for the most flexible one. ERCOT, which is almost islanded, receives $\tau = 0.95$ ($u_i = 19$); the Chelan County PUD in the densely interconnected Pacific Northwest receives $\tau = 0.80$ ($u_i = 4$). This calibration is deliberately simple and transparent; Section 6.7 shows that the ranking of methods is the same under common critical ratios between 0.5 and 0.95.

6.3 Candidate rules

Seven candidate forecasts are produced for every BA, day and hour, each using only information available before the operating day: (1) the BA’s official day-ahead forecast; (2) persistence (same hour of $d-1$); (3) weekly persistence (same hour of $d-7$); (4) the mean of the same hour over $d-1, \dots, d-7$; (5) the mean of the same hour on $d-7$, $d-14$ and $d-21$; (6) a BA-and-hour-specific linear regression on the $d-1$ and $d-7$ values, the official forecast and weekday dummies, estimated on a rolling 56-day window and refitted weekly; and (7) a single global LightGBM model ([Ke et al., 2017](#)) across all BAs and hours, trained on lags, calendar variables and the official forecast, all scaled by the BA’s recent mean demand, refitted every 28 days on the preceding 365 days. Model (7) is the kind of machine-learning forecaster a central planner could run for every BA at low cost. Each forecast $f^{(m)}$ becomes a candidate commitment by

adding its own safety margin,

$$q_{i,d,h}^{(m)} = f_{i,d,h}^{(m)} \{1 + \hat{Q}_{\tau_i}(e_i^{(m)})\},$$

where $\hat{Q}_{\tau_i}(e_i^{(m)})$ is the empirical τ_i -quantile of the relative errors $(y - f^{(m)})/\max\{f^{(m)}, 0.2\bar{y}_i\}$ of forecast m for BA i over the previous 28 days; the floor at 20% of the BA’s mean demand $\bar{y}_i$ prevents an occasional implausibly small forecast from producing an unbounded margin. The candidate rules are available from February 2020, so the evaluation period runs from 1 April 2020 to 26 September 2026: 2,342 days, 335 weekly re-estimations and 2.58 billion MWh of demand.

6.4 Competing methods

Every seven days each method re-estimates its combination from the most recent W days, $W \in \{7, 14, 28, 56\}$ (each day contributes 24 hourly observations per BA) and commits for the following seven days. Besides GDMA and soft GDMA ($G_{\max} = 6$) we consider: the official-forecast policy (candidate 1 alone), which is the status quo; equal weights over the seven candidates; the best single candidate per BA over the window; pooled decision-focused weights; per-BA decision-focused weights (the decision-focused analogue of per-series forward validation); shrinkage of per-BA weights towards pooled weights with the shrinkage intensity chosen on the hold-out (a decision-focused version of [Diebold and Shin, 2019](#)); two-step clustering, which runs k -means on the per-BA weight vectors and re-estimates pooled weights within clusters, with the number of clusters chosen by the same hold-out; and two *forecast-then-commit* methods, which estimate per-BA or grouped combination weights by minimising scaled squared forecast error and then add the τ_i -quantile of the combined forecast’s relative errors. The last two isolate the value of estimating the combination for the decision rather than for accuracy.

6.5 Results

Table 3 contains the main result. Four findings stand out.

First, deciding for the decision pays. Soft GDMA lowers the system-wide commitment cost by 18–20% relative to the status quo of committing on the official forecasts, and every decision-focused combination method lowers it by at least 15%; all reductions are significant at the 1% level. The forecast-then-commit methods, which combine the same seven forecasts for accuracy and then add a quantile margin, save only 3–18%, and they are significantly more expensive than soft GDMA for every window length. The gap is largest when windows are short ($W = 7$: 0.971 and 0.907 against 0.817), because accuracy-optimal weights are chosen without regard to the asymmetric costs and the margin estimated from one week of combined errors is noisy. Equal weights are the worst choice: they give weight to the naive calendar rules and raise cost by 53%.

Second, neither extreme is right. Per-BA weights are expensive when windows are short (0.850 at $W = 7$, 4.2% above pooled weights), pooled weights are expensive when windows are long (0.804 at $W = 56$, 0.9% above soft GDMA), and the per-window ranking of the two reverses between $W = 14$ and $W = 28$. This is the bias–variance trade-off of Theorem 1 in action.

Table 3: Out-of-sample system-wide commitment cost, 1 April 2020 to 26 September 2026, relative to committing on the official day-ahead forecasts. W is the length of the estimation window. Bold: lowest cost in the column. Stars: Diebold–Mariano test of equal daily system cost against soft GDMA with Newey–West variance (7 lags); *, **, ***: $p < 0.10, 0.05, 0.01$.

Method	System cost relative to the official-forecast policy				Share of BAs improved ^a
	$W = 7$ days	$W = 14$ days	$W = 28$ days	$W = 56$ days	
Official forecast	1.0000***	1.0000***	1.0000***	1.0000***	0.00
Equal weights	1.5287***	1.5287***	1.5287***	1.5287***	0.13
Best single (per BA)	0.8490***	0.8288***	0.8245***	0.8234***	0.76
Pooled DF weights	0.8155	0.8110***	0.8075***	0.8039***	0.78
Per-BA DF weights	0.8496***	0.8221***	0.8069***	0.8007***	0.78
Shrinkage to pooled	0.8154	0.8053	0.8009	0.7962	0.80
Two-step k -means	0.8242	0.8077*	0.8041***	0.7980	0.83
Forecast-then-commit, per BA	0.9713***	0.8850***	0.8382***	0.8233***	0.65
Forecast-then-commit, grouped	0.9068***	0.8523***	0.8260***	0.8277***	0.67
GDMA	0.8162	0.8077***	0.8042***	0.7999***	0.78
GDMA, minimum hold-out rule	0.8227***	0.8108***	0.8032***	0.7977*	0.80
Soft GDMA	0.8163	0.8048	0.7999	0.7965	0.80

^a Share of the 46 BAs whose own cost is lower than under the official-forecast policy ($W = 28$).

Third, soft GDMA is the most reliable method. It has the lowest system cost for $W = 14$ and $W = 28$ and is within 0.1% of the best method for $W = 7$ and $W = 56$; no method is significantly cheaper than soft GDMA for any window, while it is significantly cheaper than pooled weights, per-BA weights, the forecast-then-commit methods and plain GDMA for most windows. Its closest competitor is decision-focused shrinkage towards pooled weights, which is the special case $G = 1$ of soft GDMA and is statistically indistinguishable from it; the two coincide exactly in the 57–68% of weeks (depending on W) in which the hold-out rule selects a single segment. Plain GDMA improves significantly on pooled weights for $W \geq 14$ and on per-BA weights for $W \leq 14$, but forcing all members of a segment onto identical weights costs 0.4–0.5% relative to soft segments when $W \geq 14$. Choosing G by the plain hold-out minimum instead of the one-standard-error rule is worse for short windows (0.823 against 0.816 at $W = 7$), confirming that parsimony matters when hold-out costs are noisy.

Fourth, the savings are economically large. At $W = 28$ the official-forecast policy costs 0.0761 units of idle-capacity cost per MWh of demand and soft GDMA 0.0609, a saving of 0.0152 units per MWh. Over the 4.0 billion MWh of annual demand in our panel, and valuing idle committed capacity at an illustrative \$10 per MWh, this is roughly \$600 million a year for the system as a whole, obtained by re-weighting forecasts that the operators already produce. The saving is not driven by a few operators: 80% of BAs are individually better off than under their own official forecast (last column of Table 3).

Table 4 and Figure 1 confirm the picture. With every BA weighted equally rather than by its size, soft GDMA and shrinkage are again the two cheapest methods, so the results are not an artefact of the three largest ISOs. The decision-focused methods hit the target shortfall frequency closely (11.5–12.0% against a target of 12.5%, the average of $1 - \tau_i$), whereas the official-forecast policy and the forecast-then-commit methods fall short too often (13.5–14.5%). The cumulative saving builds up steadily over six years rather than in a few episodes, and the

Table 4: Secondary metrics for $W = 28$ days.

Method	Scale-free cost ^b (rel. to EW)	Shortfall frequency	DM statistic vs soft GDMA ^c
Official forecast	0.7435	0.135	27.51
Equal weights	1.0000	0.083	35.98
Best single (per BA)	0.6326	0.138	7.66
Pooled DF weights	0.6262	0.117	5.34
Per-BA DF weights	0.6216	0.116	3.81
Shrinkage to pooled	0.6187	0.115	1.21
Two-step k -means	0.6218	0.120	2.90
Forecast-then-commit, per BA	0.6363	0.145	6.93
Forecast-then-commit, grouped	0.6346	0.137	7.29
GDMA	0.6244	0.119	4.53
GDMA, minimum hold-out rule	0.6225	0.120	4.62
Soft GDMA	0.6190	0.116	—
Target (mean τ_i across BAs)		0.125	

^b Each BA’s cost divided by its mean hourly demand, averaged over BAs with equal weight, relative to equal weights. ^c Diebold–Mariano statistic; positive values: the method has higher daily system cost than soft GDMA.

ordering of the methods is stable after the first year.

6.6 Segments

GDMA uses one segment in about two thirds of the weeks and two or more in the rest; multi-segment weeks occur in every year from 2021 onwards (Figure 2, left). The co-membership matrix (right) reveals two stable segments. The first contains 17 BAs, among them PJM, CAISO, ISO New England, Florida Power & Light, Arizona Public Service, Nevada Power and both Pacific-Corp areas; their combinations rely almost entirely on the learned models (weights of 0.38 on the hourly regression and 0.47 on LightGBM) and put only 0.12 directly on the official forecast. The second contains 28 BAs, including MISO, ERCOT, the Southern Company, NYISO, TVA, Bonneville and Duke Energy, and puts 0.39 directly on the official forecast. Southwest Power Pool often forms a segment of its own. The two main segments have the same average critical ratio (0.874 and 0.873), so the segmentation is not a relabelling of cost asymmetries; nor does it follow geography or market design, since both contain ISOs and vertically integrated utilities from the East and the West. What separates them is how much the operator’s own forecast adds to what can be learned from recent demand—information that a central planner cannot read off the organisational chart of the grid, but that GDMA extracts from the data.

6.7 Extreme events and robustness

Table 5 examines three extreme episodes. During Winter Storm Uri and the 2023 heat wave, the decision-focused combinations cost 15–22% less than the official-forecast policy, in line with their average performance. Winter Storm Elliott is different: the sudden cold snap of December 2022 was forecast much better by the operators, whose forecasts use weather information that none of our candidates sees directly, and every combination is 18–32% more expensive than the official policy over those six days. Soft GDMA and shrinkage limit the damage better than pooled weights and plain GDMA (1.21 against 1.32), presumably because unit-level shrinkage lets the

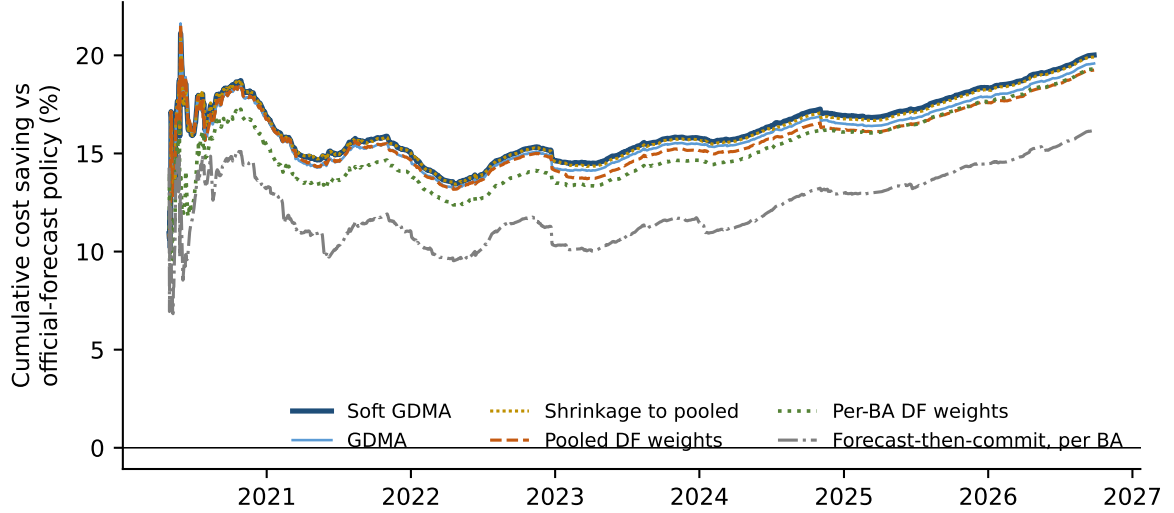


Figure 1: Cumulative system-wide cost saving relative to the official-forecast policy ($W = 28$).

Table 5: System cost during extreme events relative to the official-forecast policy ($W = 28$): Winter Storm Uri (10–20 February 2021), Winter Storm Elliott (22–27 December 2022) and the heat wave of 15 July–15 August 2023.

Method	Uri	Elliott	Heat wave 2023
Official forecast	1.000	1.000	1.000
Equal weights	1.133	2.534	1.366
Best single (per BA)	0.881	1.264	0.796
Pooled DF weights	0.842	1.318	0.778
Per-BA DF weights	0.848	1.184	0.790
Shrinkage to pooled	0.845	1.210	0.779
Two-step k -means	0.842	1.239	0.780
Forecast-then-commit, per BA	0.973	1.584	0.796
Forecast-then-commit, grouped	0.946	1.481	0.785
GDMA	0.842	1.318	0.782
GDMA, minimum hold-out rule	0.841	1.221	0.788
Soft GDMA	0.845	1.210	0.780

BAs whose official forecasts are most informative keep more weight on them. The episode is a reminder that combination weights learned in calm weeks cannot anticipate an abrupt regime change; adding weather-based candidates, or letting weights respond to weather warnings, is a natural extension.

Table 6 replaces the calibrated critical ratios by a common τ for every BA, from the symmetric case $\tau = 0.5$ to the highly asymmetric $\tau = 0.95$. The level of the savings changes little—soft GDMA saves 19–23% relative to the official-forecast policy—and so does the ranking: soft GDMA is the cheapest method for $\tau \in \{0.5, 0.9, 0.95\}$ and under the calibrated ratios, and ties with shrinkage for $\tau = 0.8$. The advantage of decision-focused over forecast-then-commit combination grows with the asymmetry, from 2.5% at $\tau = 0.5$ to 3.4% at $\tau = 0.95$ relative to the better forecast-then-commit method, as expected: the more asymmetric the cost, the more the best combination for the decision departs from the most accurate one.

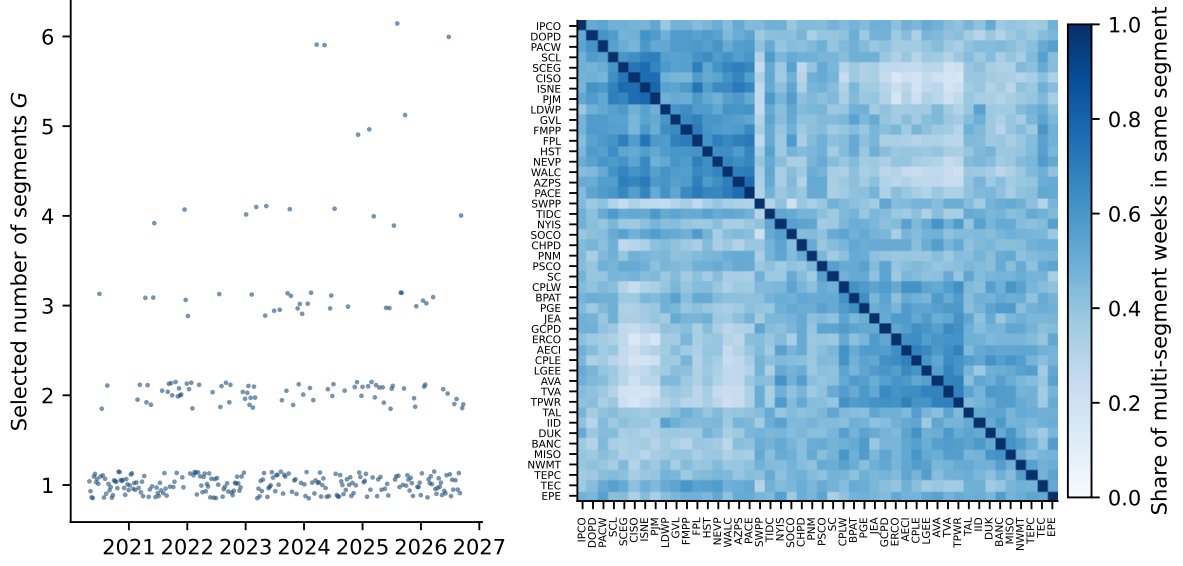


Figure 2: Left: number of segments selected by GDMA in each weekly re-estimation ($W = 28$). Right: among the weeks with $G \geq 2$, share of weeks in which two BAs belong to the same segment, ordered by average-linkage clustering.

7 Capacity commitment on five continents

7.1 A global panel of grid areas

We assemble public hourly demand data for 125 grid areas on five continents (Table 7). *North America* is represented by the 46 U.S. balancing authorities of Section 6 (EIA Form 930, 2019–2026), the only systems in the panel that publish their own day-ahead forecasts. *Europe* contributes 33 national systems from the Energy-Charts platform of Fraunhofer ISE (Fraunhofer Institute for Solar Energy Systems ISE, 2026), 2018–2026, including cross-border trading flows. *Oceania* contributes the five regions of Australia’s National Electricity Market (Australian Energy Market Operator, 2026), 2018–2026. *Asia* contributes the 31 provinces of mainland China for 2018 (Wu and Kan, 2023), the four grid areas of Taiwan for 2018 to mid-2022 (Fu and Chen, 2023) and the five regions of Thailand for 2023–2024 (Bunnak, 2025). *Africa* is represented by Algeria (Sonelgaz), 2018 to January 2020 (Rania, 2023). All series are converted to hourly UTC, cleaned as in Section 6, and placed on a common calendar from 1 January 2018 to 26 September 2026. Together the areas have an average demand of 1.66 TW—North America 455 GW, Europe 339 GW, Asia 839 GW, Oceania 21 GW and Africa 7 GW—roughly half of world electricity demand, although not all areas are observed at the same time.

The panel is deliberately unbalanced. Systems in Asia and Africa enter and leave, and several have only one or two years of data, which is typical of the data situation in emerging economies; an area contributes to estimation and evaluation only on days with data. Public hourly data for other large African and Asian systems (South Africa, India, Japan, South Korea) exist but could not be retrieved without registration or were not reachable from our computing environment; we discuss this limitation in Section 9.

Table 6: Sensitivity to the cost asymmetry ($W = 28$): system cost relative to the official-forecast policy when every BA has the same critical ratio τ , and under the calibrated heterogeneous ratios.

Method	$\tau = 0.5$	$\tau = 0.8$	$\tau = 0.9$	$\tau = 0.95$	Calibrated
Official forecast	1.0000	1.0000	1.0000	1.0000	1.0000
Equal weights	1.4033	1.4353	1.4860	1.5265	1.5287
Best single (per BA)	0.8158	0.7921	0.8081	0.8326	0.8245
Pooled DF weights	0.7988	0.7775	0.7897	0.8136	0.8075
Per-BA DF weights	0.8007	0.7798	0.7914	0.8140	0.8069
Shrinkage to pooled	0.7932	0.7714	0.7834	0.8067	0.8009
Two-step k -means	0.7950	0.7736	0.7857	0.8101	0.8041
Forecast-then-commit, per BA	0.8136	0.7944	0.8144	0.8483	0.8382
Forecast-then-commit, grouped	0.8128	0.7916	0.8066	0.8337	0.8260
GDMA	0.7968	0.7750	0.7868	0.8094	0.8042
GDMA, minimum hold-out rule	0.7981	0.7762	0.7880	0.8098	0.8032
Soft GDMA	0.7931	0.7714	0.7832	0.8065	0.7999

Table 7: The global panel.

Continent	Systems	Areas	Period	Mean demand
North America	U.S. balancing authorities (EIA-930)	46	2019–2026	455 GW
Europe	National systems (Energy-Charts)	33	2018–2026	339 GW
Oceania	NEM regions, Australia (AEMO)	5	2018–2026	21 GW
Asia	China (31 provinces)	31	2018	
	Taiwan (4 grid areas)	4	2018–2022	839 GW
	Thailand (5 regions)	5	2023–2024	
Africa	Algeria (Sonelgaz)	1	2018–2020	7 GW
Total		125	2018–2026	1.66 TW

7.2 Candidates, costs and competitors

Every area receives eight candidate forecasts: the seven of Section 6 and a zero-shot forecast from the pretrained time-series foundation model Chronos-Bolt (Ansari et al., 2024), which receives the previous 28 days of hourly demand and returns a 24-hour forecast without any fine-tuning. The LightGBM model is trained jointly on all 125 areas with continent and area identifiers and is refitted every 56 days on the preceding year. Outside the United States no official forecast exists; there the “official” candidate is set equal to the LightGBM forecast, which leaves the set of attainable combinations unchanged, and the benchmark policy is therefore the *single best forecaster*: the operator’s forecast where one is published and the global LightGBM model elsewhere. Over the evaluation period the mean absolute errors, scaled by mean demand, are 3.6% for LightGBM, 3.8% for Chronos and for the hourly regression, 4.5% for the official/LightGBM candidate and 6–8% for the naive rules; the foundation model is thus competitive with a model trained on the data themselves.

Critical ratios follow the flexibility rule of Section 6 wherever interchange is measured (the United States, 31 European countries, and the isolated island grid of Taiwan, whose flexibility is zero); the remaining areas receive the midpoint $\tau = 0.875$. Section 6.7 and Table 14 in the appendix show that a common τ gives the same ranking.

Besides the benchmarks of Section 6 we add a deep-learning competitor, a *decision-focused neural gate*. It is a two-hidden-layer perceptron (6,920 parameters) that maps each area’s fea-

Table 8: Out-of-sample commitment cost on five continents ($W = 28$ days, May 2018 to September 2026), relative to the single best forecaster. Bold: lowest in the column. Stars: Diebold–Mariano test of equal daily cost against soft GDMA (Newey–West, 7 lags); *, **, ***: $p < 0.10, 0.05, 0.01$.

Method	World	North America	Europe	Oceania	Asia	Africa
Single best forecaster ^a	1.000***	1.000***	1.000***	1.000***	1.000***	1.000***
Equal weights	1.315***	1.472***	1.183***	1.023***	1.175***	1.039***
Best single (per area)	0.870***	0.851***	0.890***	0.861	0.889***	0.873***
Pooled DF weights	0.872***	0.847***	0.898***	0.926***	0.864***	0.826*
Per-area DF weights	0.838***	0.806	0.867***	0.857**	0.879***	0.831**
Shrinkage to pooled	0.833***	0.804	0.861***	0.868***	0.852***	0.813
Two-step k -means	0.832***	0.810***	0.851	0.862	0.855***	0.824
Forecast-then-commit, per area	0.923*	0.936	0.880***	0.857*	1.124	0.840
Forecast-then-commit, grouped	0.917	0.950	0.852	0.855***	1.098	0.817
Neural gate (deep MoE)	0.918***	0.942***	0.889***	0.852***	0.954***	0.841***
GDMA	0.835***	0.816***	0.852	0.870***	0.849	0.811
Soft GDMA	0.828	0.804	0.850	0.861	0.847	0.810
Grid areas	125	46	33	5	40	1

^a The operator’s official forecast where one is published (United States), otherwise the global LightGBM forecast.

tures at the weekly origin—each candidate’s recent decision cost and forecast error over the previous 7 and 28 days, the area’s critical ratio, size and continent, and the season—into combination weights through a softmax, as in mixture-of-experts model averaging (Gong et al., 2026). It is trained end to end on the same smoothed newsvendor cost, on the realised outcomes of the previous 52 weeks, and re-trained every four weeks. Unlike GDMA, it can let weights vary continuously with observed conditions; like GDMA, it only uses information available at the time of each decision.

7.3 Results

Table 8 shows that the results of the U.S. analysis hold on every continent. Worldwide, soft GDMA lowers the commitment cost by 17.2% relative to the single best forecaster and is significantly cheaper, at the 1% level, than every competitor except the two forecast-then-commit methods, which are 11% more expensive but whose daily costs are so volatile that the test is inconclusive; its margin over the closest competitors, two-step clustering and shrinkage, is small (0.4–0.6%) but statistically clear. It has the lowest cost in Europe, Asia and Africa, is statistically tied with shrinkage and per-area weights in North America, and is 1.1% more expensive than the best method in Oceania, where only five regions are available.

Three comparisons stand out. First, decision-focused combination again dominates forecast-then-commit: combining the same eight forecasts for accuracy and adding a quantile margin saves only 8% worldwide and *raises* cost in Asia by 10–12%, where short histories make the estimated margins of the combined forecasts unreliable. Second, the neural gate is clearly worse than the simple decision-focused rules (0.918 against 0.828 worldwide) despite its flexibility; it is the best method only in Oceania. Its weights are learned from a year of past weeks and transfer poorly to weeks in which the relative performance of the candidates has shifted, whereas GDMA re-estimates a few interpretable rules every week. We did not tune the network extensively, so the comparison should be read as evidence that flexibility is not what this problem needs, not as

Table 9: Commitment cost of areas entering the panel, relative to the single best forecaster ($W = 28$).

Method	First 8 weeks after entry ^b	All other weeks
Single best forecaster ^a	1.000	1.000
Equal weights	2.120	1.307
Best single (per area)	0.994	0.869
Pooled DF weights	1.233	0.868
Per-area DF weights	0.991	0.836
Shrinkage to pooled	1.012	0.831
Two-step k -means	0.971	0.831
Forecast-then-commit, per area	1.037	0.922
Forecast-then-commit, grouped	1.283	0.914
Neural gate (deep MoE)	1.629	0.911
GDMA	1.039	0.833
Soft GDMA	0.978	0.827
Grid areas entering	52	

^b The first eight weeks after an area has at least one day of data in the estimation window.

a statement about the best achievable neural design. Third, the ranking is stable across windows (Table 13 in the appendix): with $W = 14$ soft GDMA saves 15.9% and is again the cheapest method worldwide.

New and data-poor systems. Fifty-two areas enter the panel after the start of the evaluation: the 46 U.S. balancing authorities in January 2019, the five Thai regions in January 2023 and one European system. During their first eight weeks each of them has only a few weeks of history (Table 9). Pooled weights, which impose the rule learned from other systems, are 23% more expensive than the single best forecaster in this phase, and the neural gate, whose features are not yet informative, is 63% more expensive. Two-step clustering (0.971) and soft GDMA (0.978) are the only methods that clearly beat the single best forecaster, and per-area weights are close behind (0.991). The gain from segments in this phase is thus modest; what the entry experiment shows most clearly is that a single global rule, whether pooled or learned by a network, can be costly for a system whose conditions differ from the majority.

Segments across continents. GDMA uses more than one segment in 75% of the weekly re-estimations (two segments in 42%, three in 21%), much more often than in the U.S. panel, and soft GDMA shrinks towards the segment rule with an average intensity of $\kappa = 0.70$. Segments are not continents. Among weeks with several segments, two European systems share a segment 67% of the time and two U.S. balancing authorities 53% of the time, but European, Oceanian and African systems are grouped together about 70% of the time, and North American areas are grouped with European ones only 28% of the time. The segmentation therefore does not follow geography; it is consistent with differences in which forecasting systems work—notably whether an operator’s own forecast is available and informative—which a planner could not have specified in advance.

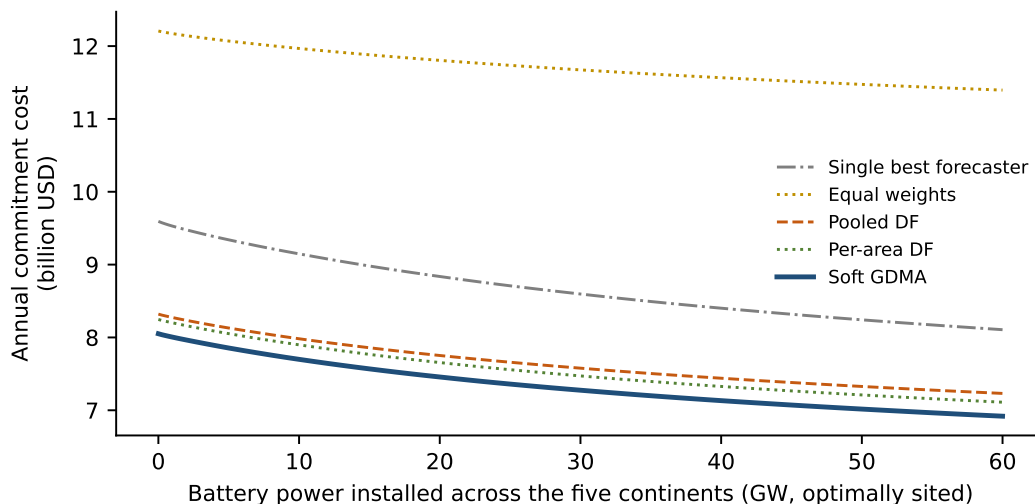


Figure 3: Annual system-wide commitment cost as a function of battery power installed and optimally sited across the 125 grid areas, for five commitment rules. Costs valued at \$10 per MWh of idle committed capacity.

8 Where to install batteries

We now solve the planning problem (4) on the global panel. For each commitment rule we take its out-of-sample hourly shortfalls and surpluses ($W = 28$), place a four-hour battery with 85% round-trip efficiency in every grid area, and compute the annual value of battery power on a grid from 0 to 15% of the area’s mean demand. Costs are converted to money by valuing one MWh of idle committed capacity at \$10, so that a shortfall costs between \$40 and \$190 per MWh depending on the area’s critical ratio; the value of battery power scales proportionally with this assumption.

8.1 Batteries do not pay for themselves on forecast errors alone

The first increment of battery power is worth \$38,000–\$46,000 per MW-year in the median area of each continent (depending on the commitment rule), and at most \$84,000 (in Great Britain). Annualised costs of four-hour batteries are well above this range (we use \$100,000–\$200,000 per MW-year), so the cost-optimal battery size for covering day-ahead commitment errors alone is zero in every area and for every commitment rule. Storage is deployed for arbitrage, capacity adequacy and the integration of renewables, and covering commitment errors is a by-product. The relevant planning question is therefore not *whether*, but *where*, to place the storage that energy-transition programmes already mandate, and how that choice depends on the way operators commit.

8.2 Better forecasting substitutes for storage

Figure 3 and Table 10 give the central managerial result. Without storage, soft GDMA commits the five continents’ capacity for \$8.05 billion a year, against \$9.59 billion for the single best forecaster. Adding batteries lowers every curve, but the curve of the single best forecaster never reaches soft GDMA’s starting point: even 60 GW of optimally sited batteries leave it at

Table 10: Storage siting on five continents. Cost of each commitment rule without storage, battery power it needs (optimally sited) to reach the cost of soft GDMA without storage, and the optimal siting of a 10 GW storage programme.

Commitment rule	Cost without	GW to match	Siting of a 10 GW budget (GW)				
	storage (\$bn/yr)	soft GDMA ^c	N. Am.	Europe	Oceania	Asia	Africa
Single best forecaster	9.59	> 60	3.7	1.9	0.4	4.0	0.0
Equal weights	12.21	> 60	1.8	2.1	1.1	5.0	0.1
Pooled DF	8.32	8	1.9	2.5	0.5	5.1	0.0
Per-area DF	8.25	6	2.8	2.6	0.5	4.0	0.0
Forecast-then-commit	8.36	7	4.3	2.5	0.4	2.7	0.1
GDMA	8.09	1	2.6	2.5	0.5	4.3	0.0
Soft GDMA	8.05	–	2.5	2.5	0.5	4.4	0.1

^c Smallest optimally sited budget, on a grid of 0.25 GW steps up to 5 GW and 1 GW steps up to 60 GW, at which the rule’s cost falls to or below that of soft GDMA without storage.

Table 11: Annual value (million USD) of storage programmes of different sizes when operations follow soft GDMA, by the commitment rule whose error profile was used to site the batteries.

Siting based on	1 GW	5 GW	10 GW	20 GW
Single best forecaster	39	181	341	582
Per-area DF	45	193	350	592
Forecast-then-commit	40	184	340	587
Soft GDMA	46	196	352	594

\$8.11 billion. Relative to the standard ways of combining forecasts, soft GDMA is worth 6 GW (per-area weights), 7 GW (forecast-then-commit) and 8 GW (pooled weights) of batteries. At \$150,000 per MW-year, 6–8 GW of batteries cost \$0.9–1.2 billion a year; the re-weighting of existing forecasts that replaces them costs essentially nothing.

8.3 The siting decision depends on the commitment rule

Table 10 also shows how a 10 GW programme should be distributed. Under soft GDMA, 2.5 GW go to North America, 2.5 GW to Europe, 0.5 GW to Oceania and 4.4 GW to Asia. If the planner instead uses the error profile of the single best forecaster, North America receives 3.7 GW, almost 50% more, because the status quo leaves U.S. operators with the largest shortfalls; the forecast-then-commit rule would send 4.3 GW there. Those batteries would be in the wrong place once operations improve.

Table 11 quantifies the cost of that mistake. When operations follow soft GDMA, a 1 GW programme sited with the status-quo error profile delivers \$39 million a year instead of \$46 million, a 15% loss; the loss is 8% at 5 GW, 3% at 10 GW and 2% at 20 GW, because large programmes eventually reach every area with sizeable shortfalls. Small, early programmes—the typical first phase of a national storage strategy—are therefore the most sensitive to the operational rule assumed at the planning stage. The implication for planners is sequencing: fix how forecasts are combined first, and site storage on the residual risk that remains.

9 Managerial implications and conclusion

Implications for system operators and planners. Four lessons emerge for organisations that coordinate many operating units, from grid operators and regulators to international bodies that fund energy-transition programmes. First, the object to be learned is the *decision rule*, not the forecast: the cost-minimising combination of forecasting systems depends on the cost asymmetry each unit faces, and combining for accuracy leaves 3–16% of the achievable saving on the table, and in data-poor systems can even raise cost. Second, planners need not choose between one rule for every system and a bespoke rule for each. A handful of data-driven segments, with each unit shrunk towards its segment’s rule, captures the heterogeneity that matters, is estimated with far less noise than bespoke rules, and produces a segmentation that can be discussed with operators—who is grouped with whom, and why. Third, more flexible is not better: a neural mixture-of-experts gate with thousands of parameters, trained on the same decision loss, was nine percentage points more expensive worldwide than soft GDMA, and much more expensive for systems with little history. Fourth, forecasting and investment are linked. Better commitment rules are worth several gigawatts of batteries, and a storage programme should be sited on the risk that remains after operations have been improved; siting it on today’s error profile misplaces a meaningful share of a small programme’s value. Because GDMA only re-weights forecasts that operators already produce, it can be adopted without replacing existing forecasting systems.

Limitations and extensions. Our coverage of Africa and parts of Asia is thin, because public hourly demand data for South Africa, India, Japan and South Korea could not be retrieved in our computing environment; the method is designed for exactly such data-poor systems, and extending the panel is a priority. The cost calibration is stylised: shortfall costs are tied to a transparent proxy of import flexibility rather than to market prices, and the single-period newsvendor abstracts from ramping, network constraints and multi-day commitments. The candidates use no weather information, which is why the operators’ forecasts beat every combination during Winter Storm Elliott; weather-driven candidates would enter GDMA without any change to the method. The storage stage takes the commitment rule as given; jointly optimising commitments and battery dispatch, and adding the arbitrage and adequacy value of storage, would make the siting problem richer. The theory treats the number of segments as given and relies on a high-level concentration condition; a formal analysis of the hold-out choice of G and κ , and of strong cross-sectional dependence through common weather shocks, is left for future work.

Conclusion. Grouped decision-focused model averaging combines three ideas that have so far developed separately—optimal model averaging, decision-focused learning and latent group structures—into one estimator with finite-sample and asymptotic guarantees, and links it to the siting of new capacity. On five continents it turns the forecasts operators already have into commitment rules that are cheaper, statistically stable and interpretable, and it tells planners where new storage is worth most once those rules are in place.

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A Proofs

Throughout, $\|\cdot\|_1$ is the ℓ_1 norm and constants c, C may change from line to line.

Covering the simplex. For $\epsilon \in (0, 1]$ there is a set $\mathcal{N}_\epsilon \subset \Delta_M$ with $|\mathcal{N}_\epsilon| \leq (3/\epsilon)^M$ such that every $\mathbf{w} \in \Delta_M$ has a point $\mathbf{w}' \in \mathcal{N}_\epsilon$ with $\|\mathbf{w} - \mathbf{w}'\|_1 \leq \epsilon$ (volumetric argument for the ℓ_1 ball, e.g. Vershynin, 2018, Ch. 4). Hence $\mathcal{N}_\epsilon^G$ is an ϵ -net of Δ_M^G in the norm $\max_g \|\mathbf{w}_g - \mathbf{w}'_g\|_1$, of cardinality at most $(3/\epsilon)^{GM}$.

Lipschitz property. For all $(\gamma, \mathbf{W})$ and $(\gamma, \mathbf{W}')$ with $\max_g \|\mathbf{w}_g - \mathbf{w}'_g\|_1 \leq \epsilon$, Assumption 1 gives $|\hat{R}(\gamma, \mathbf{W}) - \hat{R}(\gamma, \mathbf{W}')| \leq \bar{a}B\epsilon$ and $|R(\gamma, \mathbf{W}) - R(\gamma, \mathbf{W}')| \leq \bar{a}B\epsilon$.

Proof of Theorem 1

Fix $\epsilon = 1/(NT_e)$. The set $\mathcal{F} = \{1, \dots, G\}^N \times \mathcal{N}_\epsilon^G$ has cardinality at most $G^N(3NT_e)^{GM}$. By Assumption 2 and the union bound,

$$\Pr \left(\max_{(\gamma, \mathbf{W}) \in \mathcal{F}} |\hat{R}(\gamma, \mathbf{W}) - R(\gamma, \mathbf{W})| > x \right) \leq 2G^N(3NT_e)^{GM} \exp(-cNT_e x^2/L^2).$$

Setting the right-hand side equal to δ gives, with probability at least $1 - \delta$, $\max_{\mathcal{F}} |\hat{R} - R| \leq x_\delta := L\{(GM \log(3NT_e) + N \log G + \log(2/\delta))/(cNT_e)\}^{1/2}$. For an arbitrary $(\gamma, \mathbf{W}) \in \mathcal{P}_G$ let $\mathbf{W}'$ be its nearest net point; by the Lipschitz property,

$$\sup_{\mathcal{P}_G} |\hat{R} - R| \leq x_\delta + 2\bar{a}B\epsilon.$$

Let $(\gamma^*, \mathbf{W}^*)$ satisfy $R(\gamma^*, \mathbf{W}^*) \leq R_G^* + \eta$ for arbitrary $\eta > 0$. Since $(\hat{\gamma}, \hat{\mathbf{W}})$ minimises $\hat{R}$,

$$R(\hat{\gamma}, \hat{\mathbf{W}}) \leq \hat{R}(\hat{\gamma}, \hat{\mathbf{W}}) + x_\delta + 2\bar{a}B\epsilon \leq \hat{R}(\gamma^*, \mathbf{W}^*) + x_\delta + 2\bar{a}B\epsilon \leq R_G^* + \eta + 2x_\delta + 4\bar{a}B\epsilon.$$

Letting $\eta \downarrow 0$ and substituting $\epsilon = 1/(NT_e)$ gives (5). $\square$

Proof of Corollary 1

Theorem 1 with $\delta = \delta_{N,T} \rightarrow 0$ slowly (for instance $\log(1/\delta) = \log(NT_e)$) shows $R(\hat{\gamma}, \widehat{\mathbf{W}}) - R_G^* = O_p(r_{N,T})$ with $r_{N,T} = \{GM \log(NT_e)/(NT_e) + \log G/T_e + \log(NT_e)/(NT_e)\}^{1/2} \rightarrow 0$. Dividing by $R_G^* \geq \kappa$ yields $R(\hat{\gamma}, \widehat{\mathbf{W}})/R_G^* - 1 = O_p(r_{N,T}/\kappa) \rightarrow_p 0$. Because $R(\hat{\gamma}, \widehat{\mathbf{W}}) \geq R_N^*$ always and $R_G^* \geq R_N^*$, the second claim follows once we show $R_G^* = R_N^*$ under the stated structure. Let $\mathbf{w}_g^0$ be a common minimiser of R_i for all i with $\gamma_i^0 = g$ (padding with arbitrary vectors if $G_0 < G$); then $R(\gamma^0, \mathbf{W}^0) = N^{-1} \sum_i \min_{\mathbf{w}} R_i(\mathbf{w}) = R_N^*$, so $R_G^* \leq R_N^*$, and the reverse inequality holds because $\mathcal{P}_G$ is a subset of the unit-specific class. $\square$

Proof of Theorem 2

Step 1 (average consistency). Under Assumption 3(a), $R_G^* = R(\gamma^0, \mathbf{W}^0) = R_N^*$ and

$$\frac{\lambda}{N} \sum_{i=1}^N \|\widehat{\mathbf{w}}_{\hat{\gamma}_i} - \mathbf{w}_{\gamma_i^0}^0\|_2^2 \leq R(\hat{\gamma}, \widehat{\mathbf{W}}) - R_G^* = o_p(1)$$

by Theorem 1, since $\log N/T_e \rightarrow 0$ implies $\log G/T_e \rightarrow 0$ and $GM \log(NT_e)/(NT_e) \rightarrow 0$ for fixed G and M .

Step 2 (consistency of the segment weights). Fix $\eta \in (0, d/4)$. By Step 1 and Markov's inequality, the fraction of units with $\|\widehat{\mathbf{w}}_{\hat{\gamma}_i} - \mathbf{w}_{\gamma_i^0}^0\|_2 > \eta$ is $o_p(1)$. Since every true segment g contains at least πN units, with probability approaching one each g contains a unit i with $\|\widehat{\mathbf{w}}_{\hat{\gamma}_i} - \mathbf{w}_g^0\|_2 \leq \eta$; call $h(g) = \hat{\gamma}_i$. If $h(g) = h(g')$ for $g \neq g'$, the triangle inequality gives $\|\mathbf{w}_g^0 - \mathbf{w}_{g'}^0\|_2 \leq 2\eta < d$, contradicting Assumption 3(b). Thus h is injective, hence a bijection of $\{1, \dots, G\}$, and after relabelling $\max_g \|\widehat{\mathbf{w}}_g - \mathbf{w}_g^0\|_2 \leq \eta$ with probability approaching one. Since η is arbitrary, (i) follows.

Step 3 (memberships). On the event $E_\eta = \{\max_g \|\widehat{\mathbf{w}}_g - \mathbf{w}_g^0\|_2 \leq \eta\}$, unit i with $\gamma_i^0 = g$ can be assigned to $h \neq g$ only if $\widehat{R}_i(\widehat{\mathbf{w}}_h) \leq \widehat{R}_i(\widehat{\mathbf{w}}_g)$; every global minimiser assigns each unit to a segment minimising its own empirical cost, since otherwise reassigning the unit would lower (2). By Assumption 3 and the Lipschitz property (with $\|\cdot\|_1 \leq \sqrt{M}\|\cdot\|_2$),

$$R_i(\widehat{\mathbf{w}}_h) - R_i(\widehat{\mathbf{w}}_g) \geq \lambda(d - \eta)^2 - \bar{a}B\sqrt{M}\eta \geq \lambda d^2/4 =: \Delta$$

for η small enough. Hence misclassification of unit i on E_η requires $\sup_{\mathbf{w} \in \Delta_M} |\widehat{R}_i(\mathbf{w}) - R_i(\mathbf{w})| \geq \Delta/2$. Covering Δ_M by $\mathcal{N}_\epsilon$ with $\epsilon = \Delta/(8\bar{a}B)$ and applying the unit-level part of Assumption 2,

$$\Pr\left(\sup_{\mathbf{w}} |\widehat{R}_i(\mathbf{w}) - R_i(\mathbf{w})| \geq \Delta/2\right) \leq 2\left(\frac{24\bar{a}B}{\Delta}\right)^M \exp\left(-\frac{cT_e\Delta^2}{16L^2}\right) =: C_M e^{-c'T_e}.$$

Because the supremum is over all $\mathbf{w}$, the bound does not require independence between $\widehat{\mathbf{W}}$ and unit i 's data. A union bound over units gives $\Pr(\hat{\gamma} \neq \gamma^0) \leq \Pr(E_\eta^c) + NC_M e^{-c'T_e}$, which tends to zero because $\log N/T_e \rightarrow 0$. This is (ii).

Step 4 (oracle equivalence). On the event $\{\hat{\gamma} = \gamma^0\}$, $\widehat{\mathbf{W}}$ minimises $\widehat{R}(\gamma^0, \cdot)$, so $\widehat{\mathbf{W}} = \widetilde{\mathbf{W}}$ (if the minimiser is not unique, the two estimators select from the same set). By (ii) this event has probability approaching one. $\square$

B Out-of-sample risk

The theory concerns the population risk R evaluated at the estimated policy. In a rolling application the policy estimated on periods $1, \dots, T$ is applied in periods $T + 1, \dots, T + k$. If $\{z_{it}\}$ is β -mixing with coefficients $\beta(\cdot)$, a coupling argument (Yu, 1994) shows that the expected cost of the estimated policy in period $T + s$ differs from $\mathbb{E}R(\hat{\gamma}, \hat{W})$ by at most $L\beta(s)$. The bounds of Section 4 therefore also bound the out-of-sample cost up to a term that vanishes with the gap between estimation and application.

C Balancing authorities

Table 12: The 46 balancing authorities, ordered by mean hourly demand over 2019–2026. Flex.: interchange flexibility in 2019 (range between the 5th and 95th percentiles of hourly total interchange, divided by mean demand). τ_i : calibrated critical ratio.

Code	Balancing authority	Mean MW	Flex.	τ_i
PJM	PJM Interconnection, LLC	92,501	0.06	0.947
MISO	Midcontinent Independent System Operator, Inc.	74,187	0.07	0.943
ERCO	Electric Reliability Council of Texas, Inc.	49,746	0.03	0.950
SWPP	Southwest Power Pool	32,259	0.08	0.940
SOCO	Southern Company Services, Inc. - Trans	27,072	0.12	0.927
CISO	California Independent System Operator	25,318	0.29	0.900
TVA	Tennessee Valley Authority	18,636	0.09	0.937
NYIS	New York Independent System Operator	17,305	0.18	0.910
FPL	Florida Power & Light Co.	15,870	0.10	0.933
ISNE	ISO New England	13,227	0.16	0.920
DUK	Duke Energy Carolinas	12,210	0.23	0.907
CPLE	Duke Energy Progress East	7,044	0.37	0.880
BPAT	Bonneville Power Administration	6,631	0.84	0.827
PACE	PacifiCorp East	5,803	0.42	0.877
PSCO	Public Service Company of Colorado	5,312	0.11	0.930
NEVP	Nevada Power Company	4,522	0.33	0.883
LGEE	Louisville Gas and Electric Company and Kentucky Utilities Company	4,166	0.17	0.913
AZPS	Arizona Public Service Company	3,893	0.67	0.843
SC	South Carolina Public Service Authority	2,901	0.32	0.890
LDWP	Los Angeles Department of Water and Power	2,861	0.30	0.897
SCEG	Dominion Energy South Carolina, Inc.	2,814	0.15	0.923
AECI	Associated Electric Cooperative, Inc.	2,658	0.52	0.867
TEC	Tampa Electric Company	2,530	0.26	0.903
PGE	Portland General Electric Company	2,523	0.53	0.863
PACW	PacifiCorp West	2,392	0.55	0.860
FMPP	Florida Municipal Power Pool	2,135	0.17	0.917
IPCO	Idaho Power Company	2,097	0.79	0.833
BANC	Balancing Authority of Northern California	1,993	0.59	0.853
PNM	Public Service Company of New Mexico	1,645	0.64	0.847
TEPC	Tucson Electric Power	1,590	0.71	0.837
JEA	JEA	1,532	0.32	0.893
AVA	Avista Corporation	1,460	0.70	0.840
NWMT	NorthWestern Corporation	1,358	1.07	0.810
SCL	Seattle City Light	1,076	0.92	0.817
WALC	Western Area Power Administration - Desert Southwest Region	1,045	0.95	0.813
EPE	El Paso Electric Company	1,029	0.46	0.873
GCPD	Public Utility District No. 2 of Grant County, Washington	696	1.58	0.807
CPLW	Duke Energy Progress West	564	0.61	0.850
TPWR	City of Tacoma, Department of Public Utilities, Light Division	540	0.56	0.857
IID	Imperial Irrigation District	441	0.91	0.820
TIDC	Turlock Irrigation District	328	0.81	0.830
TAL	City of Tallahassee	322	0.32	0.887
DOPD	PUD No. 1 of Douglas County	245	2.20	0.803
GVL	Gainesville Regional Utilities	240	0.50	0.870
CHPD	Public Utility District No. 1 of Chelan County	218	4.70	0.800
HST	City of Homestead	71	0.90	0.823

D Additional results for the global panel

Table 13: Global panel: system cost relative to the single best forecaster for estimation windows of 14 and 28 days (world).

Method	$W = 14$	$W = 28$
Single best forecaster ^a	1.000	1.000
Equal weights	1.315	1.315
Best single (per area)	0.887	0.870
Pooled DF weights	0.874	0.872
Per-area DF weights	0.857	0.838
Shrinkage to pooled	0.846	0.833
Two-step k -means	0.843	0.832
Forecast-then-commit, per area	0.973	0.923
Forecast-then-commit, grouped	0.948	0.917
Neural gate (deep MoE)	0.918	0.918
GDMA	0.849	0.835
Soft GDMA	0.841	0.828

Table 14: Global panel with a common critical ratio $\tau = 0.9$ for every grid area ($W = 28$): system cost relative to the single best forecaster. The neural gate was not re-run for this specification.

Method	World	North America	Europe	Oceania	Asia	Africa
Single best forecaster ^a	1.000	1.000	1.000	1.000	1.000	1.000
Equal weights	1.285	1.434	1.171	1.027	1.164	1.032
Best single (per area)	0.862	0.834	0.889	0.860	0.888	0.878
Pooled DF weights	0.865	0.834	0.893	0.920	0.864	0.828
Per-area DF weights	0.832	0.788	0.869	0.856	0.886	0.839
Shrinkage to pooled	0.828	0.787	0.863	0.865	0.857	0.817
Two-step k -means	0.826	0.793	0.853	0.858	0.857	0.837
Forecast-then-commit, per area	0.894	0.887	0.880	0.854	1.039	0.846
Forecast-then-commit, grouped	0.888	0.903	0.855	0.853	1.006	0.819
GDMA	0.828	0.796	0.854	0.868	0.859	0.812
Soft GDMA	0.822	0.786	0.853	0.859	0.857	0.810